Assignment: Image Classification using CNN

1. Assignment Objective

Develop an image classification application using a **Convolutional Neural Network (CNN)** to classify images into different categories.

2. Requirements

You must:

1. Select a suitable image classification problem.
2. Collect or use a suitable image dataset.
3. Organize the dataset into **Training** and **Testing** sets.
4. Perform image preprocessing and normalization.
5. Build a CNN model using:
    - Convolutional Layer
    - Pooling Layer
    - Flatten Layer
    - Dense Layer
6. Train and evaluate the CNN model.
7. Save the trained model.
8. Develop a simple **Streamlit UI** for image prediction.
9. Test the application using new images.

3. Suggested Project Topics

You may choose one of the following:

- Cat vs Dog Classification
- Apple vs Orange Classification
- Flower Classification
- Fruit Classification
- Vehicle Classification

Assignment: Image Classification using CNN

- Animal Classification
- Food Image Classification
- Rock-Paper-Scissors Classification

You may also propose another suitable image classification problem.

4. CNN Model Requirements

The model must include:

- At least **2 Convolutional Layers**
- Pooling Layer(s)
- Flatten Layer
- At least **1 Dense Layer**
- Appropriate output layer and activation function

Students should choose suitable parameters such as:

- Image size
- Number of filters
- Kernel size
- Pool size
- Batch size
- Number of epochs
- Learning rate

5. Streamlit Application

The application should allow users to:

- Upload an image
- Display the uploaded image
- Predict the image class
- Display the predicted class

- Display prediction confidence

6. Submission Requirements

You must submit zip file include:

- Python source code
- Trained CNN model
- Dataset or dataset source
- requirements.txt
- Streamlit application
- Short project report

7. Suggested Project Title

Image Classification using Convolutional Neural Network (CNN)